

ES446 – Assignment 1

Numerical Solution of PDEs and Wave Phenomena

Instructions:

- Ideally, complete this assignment using a Jupyter Notebook.
- All plots, animations, and numerical experiments should be generated within the notebook.
- You may use **numpy**, **matplotlib**, and **ipywidgets**.
- Clearly comment your code and label all figures (axes, titles, legends where appropriate).
- Handwritten derivations for Question 3 must be scanned or neatly written and submitted as a PDF.

Reference material for Python implementations of ODEs/PDEs:

- <https://www.youtube.com/watch?v=aojkrNYfHrY>
- <https://www.youtube.com/watch?v=MM3cBamj1Ms>

Question 1

Consider the one-dimensional heat equation

1. Choose **one Dirichlet boundary condition** and **one Neumann boundary condition** (your choice).
2. Specify a reasonable initial condition.
3. Solve the equation numerically in python.
4. Plot the temperature distribution $u(x,t)$ at different times.

Ideally your plot should have sliders to change the values of the boundary condition and see how the result changes.

Question 2

Implement the two 2D PDE problems discussed in class (as presented in Tannehill et al.), using the boundary conditions exactly as given in the textbook.

(b) Right-hand problem (transient)

Solve the transient form of the PDE.

Implement a Matplotlib animation that shows how the transient response decays with time.

Clearly demonstrate how the system approaches its steady-state solution.

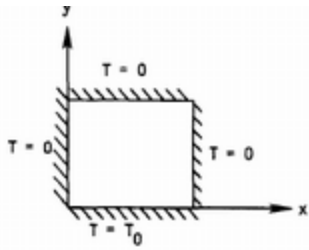
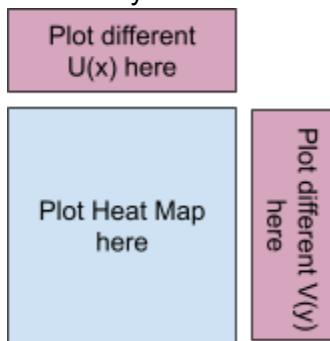


Figure 1

For this problem:

- Solve the governing PDE numerically.
- Structure your visualization as follows:



- A 2D contour or surface plot of the solution
- Clear indication of boundary conditions
- Proper axis labeling and color bar

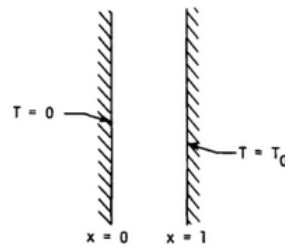


Figure 2

For this problem:

- Solve the transient form of the PDE.
- Implement a **Matplotlib animation** that shows how the transient response decays with time.
- Clearly demonstrate how the system approaches its steady-state solution.

Question 3

Go through the derivations of the above problems in detail and reproduce them by hand. Additionally, change the boundary conditions and think about how the resulting plots would change.

Then take your code in Question 2 and see if the outcome matches your expectations.

Question 4

The function

$$u(x, t) = \cos(kx - \omega t)$$

is a solution of the one-dimensional wave equation.

1. Prove this **by direct substitution** into the wave equation.
2. Animate $\cos(kx - \omega t)$ to show how the wave propagates in space.
3. Modify the solution to make the wave travel in the **opposite direction**.
4. Consider the superposition:

$$\cos(kx - \omega_1 t) + \cos(kx - \omega_2 t)$$

where $\omega_1 \neq \omega_2$.

- Animate this solution.
- Describe the resulting behavior and interpret it physically.